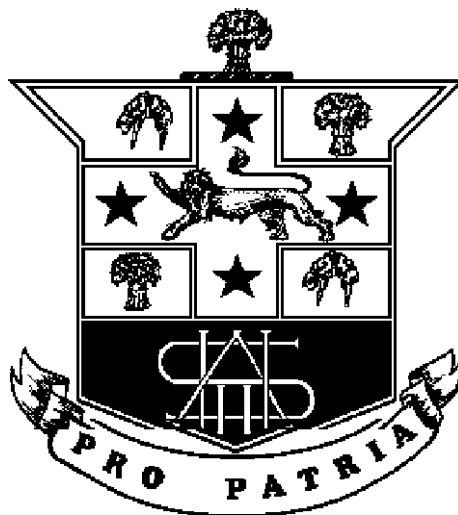




HURLSTONE AGRICULTURAL HIGH SCHOOL

YEAR 12 2012

HSC COURSE EXTENSION 1 MATHEMATICS

ASSESSMENT TASK 2 HALF YEARLY EXAMINATION

Examiners : D. Crancher, G. Rawson, S. Hackett

General Instructions

- Reading time : 5 minutes
- Working time : 90 minutes
- This exam paper has three questions. Attempt all questions.
 - Question 1 is worth 18 marks - three multiple choice questions worth one mark each and free response questions worth 15 marks.
 - Question 2 and question 3 are worth 17 marks each - two multiple choice questions worth one mark each and free response questions worth 15 marks.Total: 52 marks
- **Start a new answer booklet for each question making sure your student number is written at the top of each page.**
- All necessary working should be shown in each question. Marks may not be awarded for careless or badly arranged work.
- Board approved calculators and mathematical templates may be used.
- This examination paper must **not** be removed from the examination room.

Students Name: _____

Teacher: _____

QUESTION 1. *Start a new answer booklet.*

For parts a), b) and c) choose the correct answer from A, B, C or D and write your chosen answer in the answer booklet.

- a) How many real roots must the following equation have? **1**

$$x^4(x^2 - 4) + 9(x^2 - 4) = 0$$

A 1

B 2

C 4

D none

- b) If $P(x) = x^3 - 2x^2 + 9x - 2$, which of the following statements are true? **1**

I. $x - 3$ is a factor of $P(x)$

II. $x = 3$ is a root of $P(x) = 0$

III. $P(3) = 34$

IV. $P(-3) = 34$

A I only

B III only

C I and II only

D I and III only

For parts c), d), e), f), and g) show all necessary working out in the answer booklet.

- c) A, B and P are the points $(-1, 8)$, $(6, -6)$ and $(4, -2)$ respectively.
The point P divides the interval AB internally in the ratio $k:1$.
Find the value of k . 2
- d) If a , b and c are the roots of $x^3 - 3x + 2 = 0$, find $a^2 + b^2 + c^2$. 2
- e) Find the acute angle between the lines $y = 2x - 1$ and $3x - 2y = 5$.
Give your answer in radians correct to two decimal places. 2
- f) The polynomial $Q(x) = x^3 + ax + b$ has a factor of $(x + 2)$. 2
When $Q(x)$ is divided by $(x - 2)$ the remainder is 12.
Find the values of a and b .
- g) Let $f(x) = x^3 + 5x^2 + 17x - 10$. The equation $f(x) = 0$ has only one real root.
- i) Show that the root lies between 0 and 2. 3
- ii) Use one application of the 'halving the interval' method to find a smaller interval containing the root. 2
- iii) Which end of the smaller interval found in part ii) is closer to the root?
Briefly justify your answer. 2

QUESTION 2. *Start a new answer booklet.*

For parts a), b) and c) choose the correct answer from A, B, C or D and write your chosen answer in the answer booklet.

- a) If $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$,
then what is the $(k+1)^{\text{th}}$ term? **1**

A $4k^2$

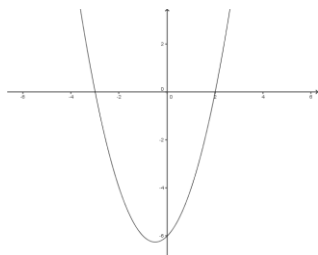
B $(2k+1)^2$

C $\frac{(k+1)2k(2k+2)}{3}$

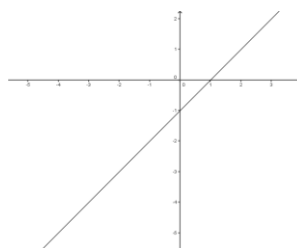
D $\frac{(k+1)(2k+1)(2k+3)}{3}$

- b) In order to solve the inequality $\frac{(x+2)(x-3)}{x+1} < 0$, which graph would be the most useful? **1**

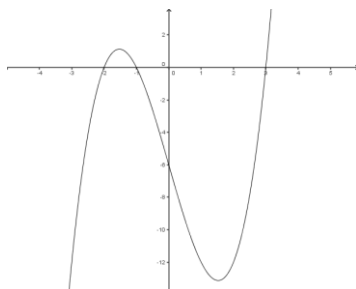
A



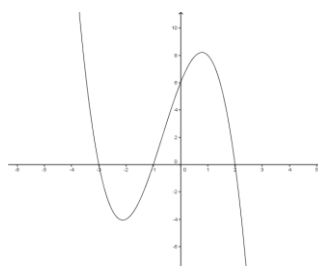
B



C



D



- c) Which of the following is NOT equal to $\sin 75^\circ$? 1

A $\sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$

B $\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$

C $\frac{\sqrt{3}+1}{\sqrt{2}}$

D $\cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ$

For parts d), e), f), g), and h) show all necessary working out in the answer booklet.

- d) Solve the inequality $\frac{x^2}{x-2} \geq -1$ 3

- e) Solve the equation $\sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$ 3

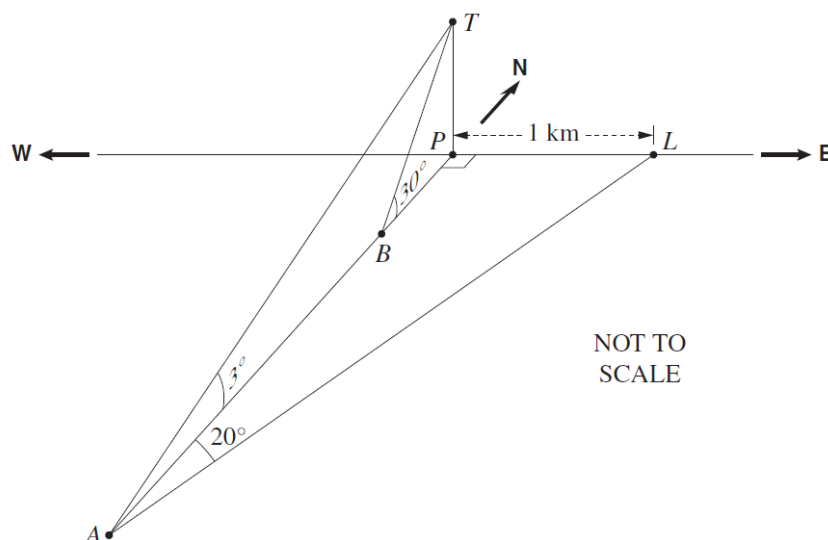
- f) Find an expression for $\sin 5x$ in terms of $\sin x$ and $\cos x$ 3

- g) Prove, by mathematical induction, that $9^{n+2} - 4^n$ is divisible by 5 for any positive integer n . 3

- h) A boat is sailing due north from a point A towards a point P on the shore line. The shore line runs from west to east.

In the diagram, T represents a tree on a cliff vertically above P , and L represents a landmark on the shore. The distance PL is 1 km.

From A the point L is on a bearing of 020° , and the angle of elevation to T is 3° . After sailing for some time the boat reaches a point B , from which the angle of elevation to T is 30° .



Show that $BP = \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$.

3

QUESTION 2. *Start a new answer booklet.*

For parts a), b) and c) choose the correct answer from A, B, C or D and write your chosen answer in the answer booklet.

- a) In a game of Poker, 5 cards are dealt to each player. **1**

The deck has 4 suits with 13 cards in each suit.

How many different hands are possible?

A ${}^{52}C_5$

B ${}^{13}C_5$

C ${}^{52}P_5$

D ${}^{13}P_5$

- b) In the same game of Poker, what is the probability of being dealt a “flush” (all five cards from the same suit)? **1**

A $\frac{5}{52}$

B $\frac{1}{52^5}$

C $\frac{33}{66640}$

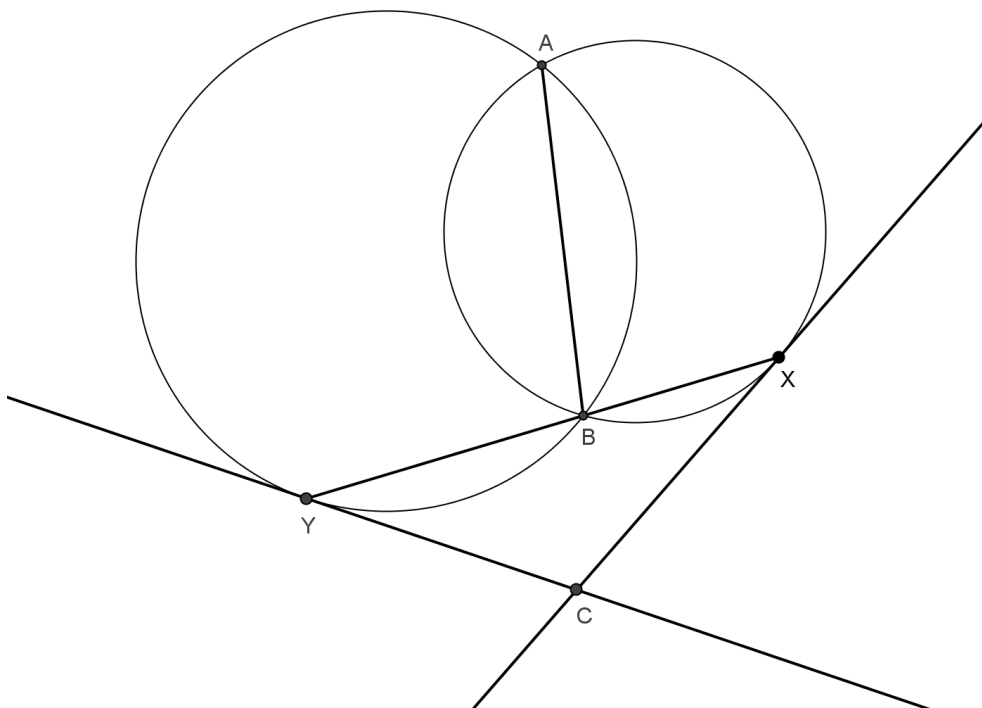
D $\left(\frac{5}{52}\right)^5$

For parts c), d), e), f), g), and h) show all necessary working out in the answer booklet.

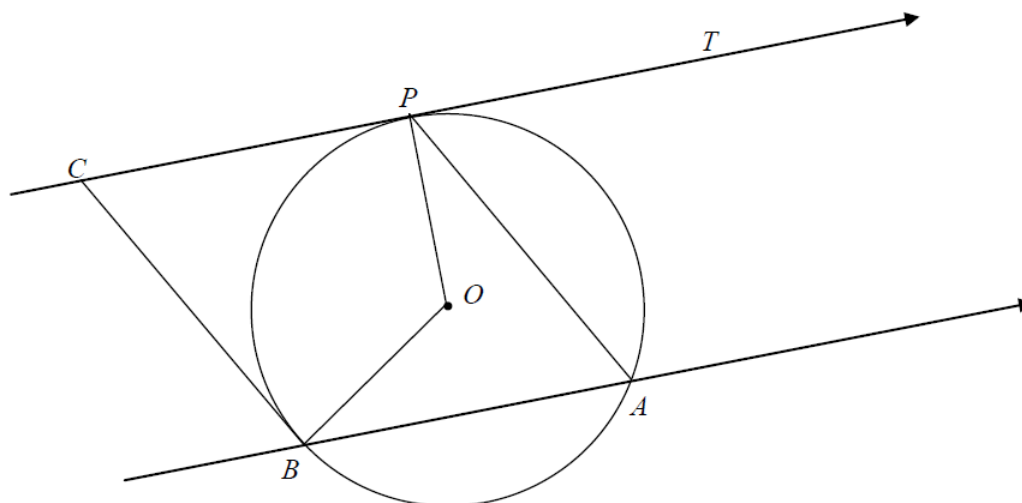
- c) AB is a common chord of two circles and a straight line through B cuts the circles in X and Y. Tangents to the circles at X and Y meet at C.

Prove that AXCY is a cyclic quadrilateral.

4



- d) CT is a tangent to the circle, centre O , touching at P . $PABC$ is a rhombus and CT is parallel to AB .



- (i) Let $\angle TPA = \alpha$ and prove that $\angle POB = 2\alpha$. 2
- (ii) Find the value of α such that $POBC$ is a cyclic quadrilateral, giving reasons. 2

- e) How many nine letter arrangements can be made using the letters of the word *ISOSCELES*? 2
- f) A club has 9 male and 7 female members. How many ways can a team of four be chosen if it is to consist of 2 men and 2 women? 2
- g) In how many ways can 6 boys and 5 girls be arranged in a row if 3 particular girls are kept together? 2
- h) Debby and John and six other people go through a doorway one at a time.
- In how many ways can the eight people go through the doorway if John goes through the doorway after Debby with no-one in between? 1

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

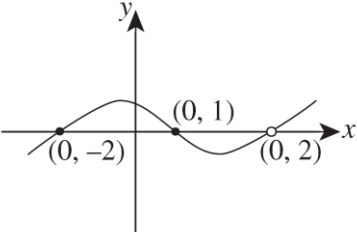
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Year 12		Extension 1 Mathematics	Task 2 HSC 2012
Question No. 1		Solutions and Marking Guidelines	
Outcome Addressed in this Question			
PE3-Solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations.			
	Solutions	Marking Guidelines	
PE3	Question 1.		
	a) $(x_1, y_1), (x_2, y_2)$ and (x, y) ratio $m : n$ $(-1, 8), (6, -6)$ and $(4, -2)$ ratio $k : 1$ $x = \frac{mx_2 + nx_1}{m + n},$ $4 = \frac{k(6) + 1(-1)}{k + 1},$ $4k + 4 = 6k - 1,$ $10 = 2k,$ $\therefore k = \frac{5}{2}$	2 marks complete correct solution 1 marks for partial correct solution	
	b) $a^2 + b^2 + c^2$ $= (a + b + c)^2 - 2(ab + bc + ac)$ $= (0)^2 - 2\left(\frac{-3}{1}\right)$ $= 6$	2 marks complete correct solution 1 marks for partial correct solution	
c) $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$ $y = 2x - 1 \text{ has gradient } m_1 = 2$ $3x - 2y = 5 \text{ has gradient } m_2 = \frac{3}{2}$ $\therefore \tan \theta = \frac{2 - \frac{3}{2}}{1 + (2) \cdot \left(\frac{3}{2}\right)}$ $= \frac{\frac{1}{2}}{4}$ $= \frac{1}{8}$ $\therefore \theta = 7^\circ 8'$ The angle between the two lines is $7^\circ 8'$	2 marks complete correct solution 1 marks for partial correct solution		

PE3	d) $Q(x) = x^3 + ax + b$ $Q(-2) = (-2)^3 + a(-2) + b = 0$ $\therefore -8 - 2a + b = 0$ $\therefore -2a + b = 8 \dots\dots\dots (A)$ $Q(2) = (2)^3 + 2(2)^2 + a(2) + b = 12$ $\therefore 8 + 2a + b = 12$ $\therefore 2a + b = 4 \dots\dots\dots (B)$ $(A) + (B)$ $2b = 12$ $\therefore b = 6$ $\therefore a = -1$	3 marks complete correct solution 2 marks partial correct solution with only one error. 1 mark for any correct work that could lead to a solution.
PE3	e) i) $f(0) = (0)^3 + 5(0)^2 + 17(0) - 10 = -10$ $f(2) = (2)^3 + 5(2)^2 + 17(2) - 10 = 52$ Since $f(0)$ is negative and $f(2)$ is positive and the polynomial is continuous between $x = 0$ and $x = 2$, then the root lies between 0 and 2.	2 marks complete correct solution 1 mark for partial correct solution
PE3	ii) $f\left(\frac{0+2}{2}\right) = f(1) = (1)^3 + 5(1)^2 + 17(1) - 10 = 13 > 0$ Hence, the roots lies between 0 and 1.	2 marks complete correct solution 1 marks for partial correct solution
PE3	iii) $f\left(\frac{0+1}{2}\right) = f(0.5) = -0.125 \text{ or } -\frac{1}{8}$ Therefore, the root lies between 0.5 and 1. The root is close to 1 than 0.	2 marks complete correct solution 1 marks for partial correct solution

Year 12	Mathematics Extension 1	H/Y Exam 2012
Question No. 2	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
HE2 - uses inductive reasoning in the construction of proofs PE3 - solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations		
Outcome	Solutions	Marking Guidelines
PE3	(a) $\frac{x^2}{x-2} \geq -1$ $x^2(x-2) \geq -(x-2)^2$ $x^2(x-2) + (x-2)^2 \geq 0$ $(x-2)(x^2 + x - 2) \geq 0$ $(x-2)(x+2)(x-1) \geq 0$  $-2 \leq x \leq 1, x > 2$	3 marks: correct solution 2 marks: Substantially correct 1 mark: Makes some progress Eg indicates $x \neq 2$ OR multiplies both sides by $(x-2)^2$ OR solves separately for $x < 2$ and $x > 2$.
	(b) $\sin 2\theta = \cos \theta$ $\sin 2\theta - \cos \theta = 0$ $2 \sin \theta \cos \theta - \cos \theta = 0$ $\cos \theta (2 \sin \theta - 1) = 0$ $\cos \theta = 0 \text{ or } \sin \theta = \frac{1}{2}$ $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$	3 marks: correct solution (must be in radians for full marks) 2 marks: Substantially correct (finds three correct solutions OR makes a minor error) 1 mark: Makes some progress (uses $\sin 2\theta = 2 \sin \theta \cos \theta$)
	(c) $\sin 3x = \sin(2x + x)$ $= \sin 2x \cos x + \cos 2x \sin x$ $= 2 \sin x \cos x \cos x + (\cos^2 x - \sin^2 x) \sin x$ $= 3 \sin x \cos^2 x - \sin^3 x$ $= 3 \sin x (1 - \sin^2 x) - \sin^3 x$ $= 3 \sin x - 4 \sin^3 x$	3 marks: for any form that includes only powers of $\sin x$ 2 marks: for incomplete expansion 1 mark: if started using any valid breakup of $\sin 3x$

HE2

(d) show true for $n = 1$.

$$\begin{aligned}9^{1+2} - 4^1 &= 729 - 4 \\&= 725 \\&= 5(145) \quad \text{which is true}\end{aligned}$$

Assume true for $n = k$

ie $9^{k+2} - 4^k = 5M$ where M is an integer

$$\text{so } 9^{k+2} = 5M + 4^k \quad \dots(1)$$

prove true for $n = k + 1$

$$\begin{aligned}9^{k+3} - 4^{k+1} &= 9 \cdot 9^{k+2} - 4 \cdot 4^k \\&= 9(4^k + 5M) - 4 \cdot 4^k \quad (\text{from (1)}) \\&= 5(4^k) + 9(5M) \\&= 5(4^k + 9M) \\&= 5N \quad \text{where } N \text{ is an integer}\end{aligned}$$

$\therefore$ true for all positive integers n by induction

$$(e) \quad \frac{PL}{PA} = \tan 20^\circ$$

$$PA = \frac{PL}{\tan 20^\circ} = \frac{1}{\tan 20^\circ}$$

$$\frac{PT}{PA} = \tan 3^\circ$$

$$PT = PA \cdot \tan 3^\circ = \frac{\tan 3^\circ}{\tan 20^\circ}$$

$$\frac{PT}{BP} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\begin{aligned}BP &= \sqrt{3} \cdot PT \\&= \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}\end{aligned}$$

3 marks: correct solution

2 marks: substantially correct

mark: partial progress towards correct solution

3 marks: correct solution

2 marks: substantially correct

1 mark: partial progress towards correct solution

Year 12 Half Yearly 2012 – Extension 1 Mathematics

Question No: 3

Solutions and Marking Guidelines

Outcomes Addressed in this Question:

PE2 - uses multi-step deductive reasoning in a variety of contexts

PE3 - solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations

Outcome	Sample Solution	Marking Guidelines
PE2	a) $\angle CXY = \angle BAX = \theta \text{ (alternate segment theorem)}$ $\angle CYX = \angle YAB = \alpha \text{ (alternate segment theorem)}$ $\alpha + \theta + \angle YCX = 180^\circ \text{ (angle sum of } \triangle CYZ = 180^\circ)$ $\therefore \angle YCX = 180^\circ - (\alpha + \theta)$ $\angle YAX + \angle YCX = \alpha + \theta + 180^\circ - (\alpha + \theta)$ $= 180^\circ$ $\therefore AYCX \text{ is cyclic (opposite angles are supplementary)}$	4 marks – correct solution with reasons 3 marks – three of four correct deductions with correct reasons 2 marks – two of four correct deductions with correct reasons 1 mark – one of four correct deductions with correct reasons
PE2	b) i) $\angle PAB = \angle TPA = \alpha \text{ (alternate angles, } PC \parallel AB)$ $\angle POB = 2\angle PAB \text{ (angle at the centre equals twice the angle at the circumference standing on arc PB)}$ $= 2\alpha$	2 marks – correct answer 1 mark – substantial progress towards correct answer
PE2	b) ii) $\angle PCB = \angle PAB = \alpha \text{ (opposite angles of a rhombus are equal)}$ $\angle PCB + \angle POB = 180^\circ \text{ (opposite angles of a cyclic quadrilateral are supplementary)}$ $2\alpha + \alpha = 180^\circ$ $\therefore \alpha = 60^\circ$	2 marks – correct answer 1 mark – substantial progress towards correct answer
PE3	c) $\frac{9!}{3!2!} = 30240$	2 marks – correct answer 1 mark – substantial progress towards correct answer
PE3	d) ${}^9C_2 \times {}^7C_2 = 756$	2 marks – correct answer 1 mark – substantial progress towards correct answer
PE3	e) $9 \times 3! = 2177280$	2 marks – correct answer 1 mark – substantial progress towards correct answer
PE3	f) $7! = 5040$	1 mark – correct answer